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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

ME1104-1 – THERMAL ENGINEERING
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. All questions carry equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **A open jar filled with mixture of water and ice is an example of which of the following?**
A) Closed System
B) Homogeneous system
C) Isolated System
D) Heterogeneous system
2. **The measurement of a thermodynamics property known as temperature is based on**
A) Zeroth law of thermodynamics
B) First law of thermodynamics
C) Second law of thermodynamics
D) None of the above
3. **Work done in a constant volume process is**
A) Negative
B) Zero
C) Positive
D) none of the above
4. **Heat transfer is**
A) point function
B) transfer function
C) path function
D) none of the above
5. **The second law of thermodynamics defines**
A) Heat and work
B) Enthalpy
C) Entropy
D) internal energy.
6. **If the temperature of the source is increased, the efficiency of the Carnot engine**
A) Decreases
B) Remains same
C) Increases
D) will be equal to the efficiency of a practical engine
7. **For a reversible adiabatic process, the change in entropy is**
A) zero
B) Minimum
C) infinite
D) unity
8. **Which of the following statements is correct according to Clausius statement of second law of thermodynamics?**
A) It is impossible to transfer heat from a body at a lower temperature to a body at a higher temperature.
B) It is impossible to transfer heat from a body at a lower temperature to a body at a higher temperature, without the aid of an external source.
C) It is possible to transfer heat from a body at a lower temperature to a body at a higher temperature by using a refrigeration cycle
D) None of the above
9. **The dryness fraction of super-heated steam is**
A) zero
B) Infinite
C) More than one
D) unity

10. Which is the prospective Thermodynamic cycle for vapor power plants?
 A) Otto cycle B) Diesel cycle
 C) Rankine cycle D) Brayton cycle
11. The efficiency of an ideal regenerative cycle is _____ the Carnot cycle efficiency
 A) More than B) Equal to
 C) Less than D) half
12. The major source of irreversibility in various components of vapor power plant is
 A) Heat loss from steam to surroundings. B) Pressure drop caused due to fluid friction
 C) Cooling of condensate below the saturation temperature in the condenser. D) All of these
13. The Brayton cycle consists of
 A) Two reversible isothermal processes and two reversible isobaric processes. B) Two reversible isochoric processes and two reversible adiabatic processes.
 C) Two reversible isothermal processes and two reversible isochoric processes. D) Two reversible isobaric processes and two reversible adiabatic processes.
14. Compare the thermal efficiency of Otto, Diesel and Dual cycles based on same compression ratio and same amount heat rejection.
 A) $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$ B) $\eta_{Dual} > \eta_{Diesel} > \eta_{Otto}$
 C) $\eta_{Otto} > \eta_{Diesel} > \eta_{Dual}$ D) $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$
15. Calculate the thermal efficiency of Brayton cycle with pressure ratio of 8.
 A) 56.47 % B) 47.5 %
 C) 44.8 % D) 50.16 %
16. The compression ratio in diesel engines is of the order
 A) 5 to 7 B) 7 to 10
 C) 7-14 D) 14-20
17. The effects of superheating the vapor in the evaporator and subcooling in the condenser of a vapor compressor refrigerator
 A) decrease of COP B) increase of COP
 C) superheating increases COP, but undercooling decreases COP D) superheating decreases COP, but undercooling increases COP
18. In a vapor compression refrigeration system, why is wet compression not preferred?
 A) the liquid refrigerant can be trapped in the head of cylinder. B) this may damage the valves or cylinder head.
 C) liquid refrigerant can wash away the lubricating oil thus accelerating wear. D) All of these
19. The most commonly used refrigerant in vapor absorption systems is
 A) Ammonia B) Freon-12
 C) Carbon dioxide D) Air
20. During sensible cooling, wet-bulb temperature
 A) decreases B) Remains same
 C) may increase or decrease D) increases

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Differentiate the following: i. Open system and closed system ii. Adiabatic system and isolated system	4	L*2	1	1
b) What do you mean by thermometer and thermometric properties? Name different thermometers and mention their thermometric properties.	4	L2	1	1

- c) A centrifugal pump delivers 50 kg of water per second. The inlet and outlet pressures are 1 bar and 4.2 bar respectively. The suction is 2.2 m below the center of the pump and delivery is 8.5 m above the center of the pump. The suction and delivery pipe diameters are 20 cm and 10 cm respectively. Determine the capacity of the electric motor to run the pump. 8 L3 2 2
2. a) A spherical balloon of diameter 0.5 m has an inside pressure of 100 kPa. Due to heating, the pressure inside the balloon increases to 400 kPa, during which the inside pressure varies as the square of the diameter of the balloon. Determine the displacement work during this process. 8 L3 1 2
- b) State first law of thermodynamics for cycle and non-cyclic process. Prove that energy is a property of the system. 6 L2 2 1
- c) Briefly explain PMM-II. 2 L2 2 1
3. a) Derive the expressions for work done and heat transferred for the following: i. Isothermal process ii. Adiabatic process 8 L2 1 1
- b) A refrigerator produces 2 tons of ice at 0°C per day from water maintained at 0°C. It rejects heat to the atmosphere at 27°C. The power to run the refrigerator is supplied by an engine which absorbs heat from a source which is maintained at 227°C by burning the fuel of calorific value 20000 kJ/kg. Find the consumption of fuel per hour and power developed the engine. Assume both devices run on Carnot cycle. 8 L3 2 1

Unit – II

4. a) Draw pressure vs volume diagram of water and indicate all the important regions. 5 L2 3 1
- b) Mention limitations of Carnot's cycle. 3 L2 3 1
- c) An engine working on Otto cycle has total volume of 0.5 m³. The pressure and temperature at the beginning of the compression stroke are 1 bar and 27°C respectively. The pressure at the end of compression stroke is 10 bar. The heat added during constant volume process is 200 kJ. Calculate the compression ratio, clearance volume, air standard efficiency and mean effective pressure. 8 L3 4 2
5. a) In steam power plant working on the Rankine cycle, steam enters the turbine with a pressure of 50 bar and a temperature of 500°C. The steam enters the condenser at a pressure of 0.05 bar. If the isentropic efficiency of the turbine and pump are 80% and 70% respectively. Determine the cycle efficiency. 8 L3 3 2
- b) Derive an expression for the air standard efficiency of the Diesel cycle. 8 L2 4 1
6. a) Explain the working of the regeneration cycle with an open feed water heater and also derive an expression for the mass of bled steam. 8 L2 3 1
- b) A gas turbine working on Brayton cycle receives air at 1 bar and 27°C. The air is compressed adiabatically to 6.2 bar with compressor efficiency being 88%. The fuel has heating value of 44180 kJ/kg and fuel air ratio being 0.017kg fuel per kg of air. If the efficiency of the turbine is 90% find: Compressor work, Turbine work and Efficiency of the cycle. 8 L3 4 2

Unit – III

7. a) Derive an expression for COP of 'Air refrigeration cycle'. 6 L2 5 1
- b) The following conditions are given for a hall to be air-conditioned: Outdoor condition: 40°C DBT and 20°C WBT. Required conditions are 20°C, 60% RH. The seating capacity of the hall is 1500. The amount of outdoor air supplied is 0.3 m³/min per person. If the required condition is achieved first by adiabatic humidification and then by cooling, estimate the capacity of the cooling coil and humidifier. 10 L3 5 1

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8. a) Explain the working of winter air conditioning with suitable diagrams. 6 L2 5
- b) A vapor compression refrigeration system of 5 ton capacity uses R-12 and operates between condenser and evaporator temperatures of 40°C and – 10°C respectively. Determine: i) Refrigerant flow rate ii) Volume flow rate handled by the compressor in m³/s iii) Compressor discharge temperature iv) Pressure ratio and COP v) Heat rejected to the condenser vi) Power required to drive the compressor. 10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
